

Dipole Atlas: theory, numerical design and interpretation

A transparent computational laboratory for separating orientational order, positional structure and molecular mobility. Six material-inspired, rigid two-site models are compared under four interaction Hamiltonians. This document explains what is calculated, why, and where inference must stop. These are classical illustrative surrogates, not validated chemical force fields or measured phase diagrams.

1. Introduction and high-level overview

1.1 Scientific question. How do interaction strength, thermal motion and imposed pressure influence molecular orientations, local packing and motion? A single dipole average can hide opposing or multi-directional populations. A single crystalline score can mistake a rotated lattice for disorder. Several independent observables are therefore retained rather than converted into an automatic phase label.

1.2 Workflow. Choose a molecular surrogate and Hamiltonian; initialize a density-controlled FCC box; minimize; equilibrate and sample each temperature plateau; export complete sampled records; then render figures and a bounded browser preview. A fresh trajectory is started for each material/potential/pressure path. Temperatures on one path inherit the previous plateau's state. The viewer selects these real precomputed trajectories; it does not rerun molecular dynamics.

1.3 Reading the interface. Potential selects the equations of motion. Diagnostic selects what is measured. Molecular display selects arrows, charge sites or centers without changing the physics. Temperature and pressure select recorded conditions independently. Smooth surfaces interpolate sampled means and are not additional simulations. Gold points remain visible to show the actual information density.

1.4 Scope. Low-volatility molecular inspirations motivate strong cohesive interactions, but no vapour pressure is measured here. Hydrogen bonding, atomistic shape, flexibility, polarization, reactions and decomposition are absent. Extreme temperatures merely stress the surrogate; real substances can decompose first. A short, tiny periodic box cannot establish melting, boiling, sublimation or coexistence temperatures.

2. Molecular geometry, units and partial charges

2.1 Each molecule contains two equal-mass sites separated by one rigid bond b . Their positions are center plus or minus $b/2$ times a unit orientation vector. The prescribed molecular mass is split equally. A rigid linear molecule has three translational and two rotational degrees of freedom; rotation about its own axis has zero moment of inertia in this point-site representation.

2.2 The positive orientation runs from the negative site to the positive site. Bond length is one quarter of the preset sigma. Charge is chosen to reproduce the illustrative dipole magnitude: one debye is 0.020819434 elementary-charge nanometers. The values approximate interaction scales, not conformer-resolved experimental dipoles. Intramolecular electrostatics and LJ are excluded because geometry is constrained.

2.3 OpenMM uses nanometers, picoseconds, daltons, elementary charges and kJ/mol. Its system energy is a molar scaling of one simulation box; dividing by the molecular count gives kJ/mol of molecules. This is a mean system energy per molecule, not a unique assignment of pair energy to each individual molecule.

2.4 The sigma scale changes the nonbonded core size, not the preset bond length or charge. The epsilon scale changes well depth independently. This permits controlled Hamiltonian comparisons without silently changing molecular geometry. The effective values and initial density are recorded in metadata. For noncharged models the orientation remains a geometric axis, but actual site charges and exported dipole magnitude are zero.

$$b = 0.25\sigma_0, \quad q/e = 0.020819434 \mu_D/b_{\text{nm}}$$

$$\mathbf{r}_{\pm} = \mathbf{R} \pm (b/2)\mathbf{u}$$

3. Explicit electrostatics, translation and rotation

3.1 All intermolecular charged sites interact through Coulomb forces. The point-dipole far-field expression is NOT used for near-field forces. In a periodic box PME includes the periodic electrostatic sum; inside walls the finite set of charges is evaluated without a spatial cutoff. These two electrostatic boundary conditions describe different systems.

3.2 Site forces automatically produce a net center-of-mass force and a torque about the center. OpenMM integrates both site coordinates with the bond constraint; there is no separate manually imposed angular relaxation rule. Unequal local electric fields on the two sites change both translational and rotational motion. The far-field dipole expression is only an asymptotic interpretation when separation is much larger than the bond length.

3.3 The molecular dipole interaction is anisotropic, but this is still a crude charge geometry. It is not equivalent to the distributed partial charges on the real atomistic molecules. The model neither includes dielectric screening by an explicit solvent nor fits a bulk dielectric constant. Bare attractive Coulomb without a repulsive core is deliberately unavailable: opposite point charges otherwise have an unbounded collapse channel.

$$U_C = \frac{1}{4\pi\epsilon_0} \sum_{a < b} \frac{q_a q_b}{r_{ab}}$$

$$\mathbf{F}_{\text{COM}} = \sum_a \mathbf{F}_a, \quad \boldsymbol{\tau} = \sum_a (\mathbf{r}_a - \mathbf{R}) \times \mathbf{F}_a$$

4. Lennard-Jones and repulsive controls

4.1 LJ combines a steep short-range repulsion and a negative attractive tail. Sigma is the zero crossing and the minimum lies at $2^{1/6}$ sigma. Site epsilon is the preset molecular epsilon times its scale, divided by four. Four intermolecular site pairs approach the nominal molecular well scale only when the bond is small relative to pair separation; this is not an exact molecular LJ potential at all orientations.

4.2 WCA keeps the LJ repulsive branch, shifts it upward by epsilon, and truncates it at its minimum. Both energy and force are continuous there. It is a useful control for separating excluded-volume packing from attraction. It is not an empirical repulsive potential fitted to any of the named materials.

4.3 Four independently simulated Hamiltonians are offered: Coulomb + LJ; LJ only; Coulomb + WCA; WCA only. LJ-only removes charges but retains cohesion. Coulomb + WCA removes dispersion attraction but retains electrostatic cohesion. WCA-only is the excluded-volume reference. Switching between them in the interface selects separate simulations, not relabeled arrows.

4.4 Periodic LJ is smoothly switched from 85% of the cutoff to zero at the cutoff and includes the OpenMM homogeneous dispersion correction. WCA has no attractive-tail correction. The neighbor-list cutoff must include the complete WCA core; incompatible choices raise an error. Periodic cutoff is $\min(\text{cutoff_sigma times effective sigma}, 0.36 \text{ times initial box length})$. These finite-box and cutoff choices require sensitivity testing before quantitative interpretation.

$$U_{\text{LJ}}(r) = 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$$

$$U_{\text{WCA}}(r) = [U_{\text{LJ}}(r) + \epsilon] \Theta(2^{1/6}\sigma - r)$$

5. Periodic boxes and stationary repulsive walls

5.1 Periodic boundaries identify opposite cube faces. Molecular bond vectors and displacements are minimum-imaged; centers are wrapped for plotting. The box has no free surface. PME tolerance is $5e-4$. Uniform dispersion corrections assume a homogeneous bulk environment and should not be trusted indiscriminately near highly inhomogeneous states.

5.2 The alternative is a finite cube with six repulsive shifted LJ wall terms applied to every site. Each wall uses its perpendicular distance d , wall sigma equal to one quarter of effective pair sigma, and an independently configurable wall epsilon. This is an external confining potential, not an integrated atomistic wall or a measured surface chemistry. The site-wall expression is the WCA equation with r replaced by d .

5.3 Walls are stationary and NVT-only. There is no imposed bulk pressure; the pressure selector explicitly says not controlled. Pair interactions in the finite system use no cutoff, no PME and no periodic tail correction. Boundary forces can bias orientations and packing. The density based on full geometric box volume differs from an accessible-volume density because the repulsive layer excludes some space.

5.4 Wall potential energy is separately exported and subtracted from total potential for the intermolecular-energy context. A site outside the box at an observation point aborts the run; this is a finite-timestep safety check, not mathematical hard-wall enforcement at every integration step. Smaller timesteps and energy/trajectory inspection remain necessary. Wall RDF is omitted because bulk spherical-shell normalization would be biased without a boundary correction.

6. Density initialization and ensemble tradeoffs

6.1 The default initial bulk density comes from attributed room-temperature reference values. These refer to 293.15 K for most materials and 297.15 K for benzyl alcohol, not exactly 300 K. Source pressure is unspecified; ambient pressure is an assumption. No invented thermal-expansion correction is applied. Formamide uses a reported relative density at 20/4 degrees C as an approximate g/cm^3 anchor.

6.2 An FCC conventional unit cell contains four molecular centers. With n cells on each edge, $N=4n^3$. The lattice parameter is calculated from molecular mass and chosen density using $1 \text{ Da/nm}^3 = 0.00166053906660 \text{ g/cm}^3$. Initial orientations are six-axis populations or seeded random unit vectors. The density override replaces the reference anchor; legacy mode retains the original sigma-based FCC spacing.

6.3 NVT keeps the chosen box volume and bulk density fixed, even as temperature changes, but pressure is an uncontrolled response and is not measured by this code. NPT initializes at the anchor, then lets the barostat change volume. One cannot independently fix N , T , P and density for a generic model. Matching a real density initially does not validate the model's equation of state or make 1 atm an equilibrium state at that density.

6.4 A positive pressure near atmospheric pressure is tiny compared with molecular repulsive stresses. Independent short paths can therefore differ mostly because of stochastic relaxation rather than a resolved pressure effect. Compare replicas, uncertainty, stationary distributions and longer trajectories before attributing a small surface feature to pressure.

$$N = 4n^3, \quad a^3 = 4M(0.00166053906660)/\rho$$

7. Integration, thermodynamic control and provenance

7.1 Energy minimization is limited to 400 iterations; it is not proof of equilibrium. Langevin-middle integration uses friction 1/ps, configured timestep (default 1 fs) and bond-constraint tolerance 1e-6. Initial velocities are sampled at the first target temperature. Temperature changes update both the integrator and, for NPT, the barostat temperature.

7.2 The periodic NPT Monte Carlo barostat proposes isotropic volume changes every 25 steps. User pressure in kPa is divided by 100 to supply bar. NVT has no barostat. Periodic systems remove center-of-mass momentum; walls do not. Consequently the temperature diagnostic uses 5N-3 degrees of freedom for periodic systems and 5N for walls. The velocity convention is that of the integrator, not an independent reconstructed kinetic ensemble.

7.3 Each material/pressure pair has a deterministic derived seed. Potential variants deliberately start with the same seed for that pair, but different forces immediately produce different paths; these are not replicas of a single Hamiltonian. Heating/cooling order is supplied by the user. Equilibration frames are excluded, displacement and rotational origins reset at each plateau, and reporting stride can differ from the integration timestep.

7.4 Metadata records configurations, effective parameters, software versions, seeds, status and SHA-256 hashes of source and theory inputs. A failed run keeps diagnostic partial records and cannot be published as complete. Existing run IDs are never overwritten. Floating-point/platform changes can prevent bitwise reproduction. Rerendering changes presentation but not the original dynamics provenance; render provenance is saved separately.

$$T_{\text{kinetic}} = \frac{2K}{fR}, \quad f = 5N - 3 \text{ (periodic)}, \quad f = 5N \text{ (walls)}$$

8. Orientational distributions and multipolar order

8.1 Equal solid angle corresponds to uniform cos(theta) and azimuth phi, not uniform theta. The canonical pooled histogram uses 10 by 16 bins; each isotropic probability is 1/160. The live preview uses fewer bins and fewer sampled molecules for speed, so its noise need not resemble the pooled map.

8.2 Polar order is the norm of the mean unit dipole. Nematic order is the largest eigenvalue of $(3 \text{mean}(\mathbf{u} \mathbf{u}^T) - \mathbf{I})/2$. Opposing axes can cancel polarization, while six equally populated coordinate directions can cancel both polar and nematic order despite a strongly nonuniform distribution. These are different symmetries, not inconsistent measurements.

8.3 The Legendre addition theorem gives rotationally invariant angular powers for degrees 1 through 6. Self-pairs are removed, making the signed estimator zero in expectation for independently isotropic directions. Negative finite-sample values remain in CSV. O6 clips each signed power at zero before averaging and taking the square root, so it has a positive finite-N noise floor. It is not the structural q6 and does not count Gaussian modes.

8.4 Pooled angular KL divergence compares histogram probabilities to the uniform distribution. Empty bins contribute zero. Finite sampling and time correlation produce a positive bias. A collectively rotating ordered configuration can have substantial instantaneous O6 but a diffuse pooled lab-frame histogram. Features above degree six can be missed; maps and spectra should be inspected together.

$$A_\ell^2 = \frac{\sum_{i \neq j} P_\ell(\mathbf{u}_i \cdot \mathbf{u}_j)}{N(N-1)}, \quad \ell = 1, \dots, 6$$

$$O_6 = \sqrt{\frac{1}{6} \sum_{\ell=1}^6 \max(0, A_\ell^2)}$$

$$D_{\text{KL}} = \sum_b p_b \ln(B p_b)$$

9. Positional order and radial structure

9.1 FCC reference order averages normalized structure-factor intensities at the starting lattice's $\{111\}$ and $\{200\}$ reciprocal vectors. It is one for the perfect reference FCC centers and approximately $1/N$ for independent uniformly random centers. Translation and uniform box scaling leave it unchanged, but lattice rotation, distortions or another crystal structure can lower it without melting. It should not be read as a universal crystalline fraction.

9.2 Local structural q6 uses the twelve closest molecular centers, with minimum imaging only in periodic systems. For each center, the degree-six Legendre polynomial is averaged over all pairs of its neighbor-direction vectors, including self-pairs; the nonnegative square root is then averaged across centers. This is rotationally invariant and equals approximately 0.574524 for ideal FCC nearest neighbors. Different neighbor definitions change the value. A fixed twelve-neighbor rule in a gas can still report nonzero order; there is no universal threshold here.

9.3 The COM radial distribution function counts unique pairs in spherical shells and normalizes by the finite-N ideal-gas expectation $N(N-1)/2$ times shell volume / box volume. Thirty-two shells extend to half the shortest box side. Shell edges are exported per frame; in NPT those edges move. Aggregate into consistent physical-r bins, or explicitly into scaled bins, rather than averaging rows as if all radii were identical.

9.4 RDF peaks reveal coordination shells; the nearest-neighbor distance tracks local spacing. Crystals, liquids and dilute fluids can differ in the extent and breadth of structure, but one peak or one scalar score alone does not identify a phase. Wall systems require boundary-corrected distributions, which are not implemented.

$$q_6(i)^2 = \frac{1}{n_i^2} \sum_{j,k=1}^{n_i} P_6(\hat{\mathbf{r}}_{ij} \cdot \hat{\mathbf{r}}_{ik})$$

$$g(r) = \frac{n_{\text{pairs}}(r)}{[N(N-1)/2] \Delta V_{\text{shell}}/V}$$

10. Translational and rotational mobility

10.1 Periodic displacements accumulate minimum-image differences of fractional COM coordinates, multiplied by the mean of neighboring box lengths. This removes affine barostat scaling. The origin resets after each plateau's equilibration. At fixed volume this is the ordinary unwrapped displacement if no reporting interval aliases a crossing by more than half the box. Sparse sampling can violate that condition even when integration itself is stable.

10.2 The displayed MSD is a single-origin mean over molecules. A linear fit to the latter half of recorded time points gives slope/6, stored as a mobility proxy. Only a long-time diffusive regime in an appropriate bulk ensemble supports the Einstein diffusion interpretation. NPT nonaffine coordinates, thermostat friction, finite-size effects and walls further complicate it. A noisy negative fitted slope is retained, not clipped or called negative physical diffusion.

10.3 Rotational C2 averages P2 of the dot product between each current axis and its post-equilibration reference axis. It is one at the origin and tends toward zero for independent isotropic orientations. Head-tail reversals are invisible to this even-parity metric. Multi-time-origin correlations, uncertainty and a demonstrable decay window are needed for a relaxation-time estimate; none is claimed by the short demo.

$$\text{MSD}(t) = \frac{1}{N} \sum_i |\Delta \mathbf{R}_i(t)|^2, \quad D_{\text{proxy}} = \frac{1}{6} \text{slope}_{\text{late}}(\text{MSD})$$

$$C_2(t) = \left\langle \frac{3[\mathbf{u}_i(t) \cdot \mathbf{u}_i(0)]^2 - 1}{2} \right\rangle_i$$

11. Fluctuation responses and phase-change diagnostics

11.1 For equilibrium NVT sampling, total-energy variance estimates C_v . For equilibrium NPT sampling, enthalpy variance estimates C_p ; volume variance gives isothermal compressibility; covariance of volume and enthalpy gives thermal expansion. Here E and H use OpenMM kJ/mol of boxes, V uses nm³, and $R=0.008314462618$ kJ/(mol K). Box enthalpy is $E + 0.000602214076 P V$ with P in kPa. Divide heat capacities by N to report per mole of molecules and multiply by 1000 for J units.

11.2 Only the ensemble-appropriate columns are populated. Wall NVT heat capacity includes external-wall energy and is a property of that confined model, not bulk material. Sample variance/covariance use a one-degree-of-freedom correction. Compressibility is converted to inverse kPa. No quantum corrections are applied. Thermostat/barostat settings, kinetic-energy conventions and constraint treatment need auditing for precision work.

11.3 Short correlated trajectories, drifting temperatures, unequilibrated densities and changing structures violate the assumptions behind fluctuation formulas. These exported values are exploratory signals, not measured material constants. Four-block standard errors for means are descriptive only; four blocks are not evidence that blocks are independent. Longer equilibration, autocorrelation analysis and independent replicas are required.

11.4 Further diagnostics worth adding: Lindemann displacement/RMSF around stable lattice sites; static structure factors and locally averaged bond order; crystalline-cluster connectivity; energy, density and order histograms with time-series stationarity; multiple-origin diffusion and rotational relaxation; hysteresis under separate heating/cooling paths; and finite-size scaling. Response peaks or bimodality can motivate further tests but are not automatically transitions.

11.5 Stronger thermodynamic evidence requires free-energy calculations or explicit two-phase coexistence, with careful interfaces, ensemble choice and finite-size analysis. A bulk periodic heating run does not measure a sublimation flux. Vapour pressure requires an appropriate coexistence/free-energy or chemical-potential method. New diagnostics should be accompanied by known-state tests, documented definitions and uncertainties rather than a universal hard-coded solid/liquid/gas threshold.

$$C_{V,m} = \frac{1000 \text{Var}(E)}{NRT^2}, \quad C_{P,m} = \frac{1000 \text{Var}(H)}{NRT^2}$$

$$\kappa_T = \frac{0.000602214076 \text{Var}(V)}{\langle V \rangle RT}, \quad \alpha_P = \frac{\text{Cov}(V, H)}{\langle V \rangle RT^2}$$

12. Empirical comparison: what can be compared?

12.1 An absolute potential-energy zero depends on the chosen Hamiltonian. There is no table of measured absolute potential energy at every selected T and P for these surrogates. The interface therefore displays Unavailable, not zero, beside the simulated energy. Density reference conditions and literature vaporization enthalpy are shown separately with sources and actual measurement conditions.

12.2 Vaporization enthalpy is an energy difference between gas and liquid plus a PV difference. If gas molecules are ideal, unassociated and internally identical to the rigid liquid molecules, and liquid PV is negligible, a rough model proxy is minus the mean intermolecular potential per molecule plus RT. It is static for the selected plateau because it uses the plateau mean, while the instantaneous potential card changes with the animation.

12.3 This proxy neglects nonideal gas interactions, gas association, flexibility, polarization, quantum effects and liquid PV. It is inapplicable as a validated vaporization enthalpy for a state not established to be liquid, especially dilute states. The viewer withholds the proxy for WCA-only controls and confined walls. Subtracting wall energy would not remove confinement from the sampled structure. A numerical resemblance to experiment would not validate the surrogate.

12.4 A reference is labeled T/P-matched only when both conditions are explicitly available and agree within 0.05 K and 0.05 kPa. Missing source pressure means no match can be claimed. No interpolation or extrapolation is applied to sparse reference data. No reference value is inferred from a boiling point without a verified enthalpy datum. The source registry and displayed tables retain missing entries.

$$\Delta H_{\text{vap}} = \Delta U + \Delta(PV)$$

$$\Delta H_{\text{vap}}^{\text{proxy}} = -\langle U_{\text{inter}} \rangle / N + RT$$

13. Surfaces, archives and reproducible visualization

13.1 Surface smoothing uses shape-preserving PCHIP first along temperature and then along log pressure. It requires a complete rectangular grid and never extrapolates. The construction is order-dependent and only interpolates the measured scalar means; it is not a fitted equation of state or phase-boundary estimator. Small apparent ridges can come from stochastic path noise. Sample-grid mode and gold markers expose the underlying resolution.

13.2 Each Hamiltonian is plotted separately, so duplicate T/P points from different models cannot overwrite one another. The chosen diagnostic controls the timeline, headline statistic and surface. The NVT viewer has no invented pressure surface. Legacy runs disable metrics that were not recorded rather than filling them with fabricated values.

13.3 Canonical records include trajectory, thermodynamics, orientation histogram and periodic RDF CSVs per condition, plus summary and metadata. The trajectory stores COM position/velocity and unit axes; the two site positions are reconstructible from the rigid bond. Angular velocities and individual site velocities are not exported. Browser previews are capped at 120 frames and 256 molecules per condition; the ZIP keeps every requested saved CSV sample.

13.4 Static plots/PDFs live in reports and evolving Canvas previews in media. All data and reports remain ordinary committable files. The nondated Physics/Dipole-Atlas/index.html route chooses the featured periodic NPT run, while archived run folders remain accessible. Query parameters encode potential, molecule, temperature, pressure, diagnostic and display without duplicating datasets. The full site generator copies pre-rendered artifacts; it never runs MD or pushes Git.

14. Design history and mandatory update checklist

14.1 Version 1 introduced explicit two-site Coulomb + LJ, periodic NPT/NVT trajectories, angular spectra, FCC reference order, CSV archives and the offline viewer. Version 2 adds six presets, four Hamiltonians, density anchors, stationary repulsive walls, q6/RDF/mobility/fluctuation diagnostics, separate selectors, honest empirical context, smooth surfaces and nondated URLs.

14.2 This JSON document is the single source for the offline mathematical HTML, printable PDF and LaTeX source. Equations are typeset as local SVGs; no CDN or TeX installation is required to read it. Parameter/reference tables are generated directly from the code registry to prevent silent table drift. The PDF is generated from the same prose and equations; a separate .tex export is provided for conventional LaTeX workflows.

14.3 When changing forces, ensembles, units, parameters, observables or reference data: update the relevant numbered section and this design history; add a numerical or invariant regression; update the README and schema if needed; regenerate the theory outputs and viewer; inspect equations and labels; preserve original run provenance; and generate a new immutable run for changed dynamics. Do not merely rerender old trajectories and claim they use the new Hamiltonian.

14.4 Before interpretation or publication: verify finite values, constraints, force signs, temperatures and volume stationarity; inspect cutoff/timestep sensitivity and sampling aliasing; check all potential/metric deep links and offline assets; confirm reference T/P labels and missing data; audit archive sizes against GitHub limits; and distinguish a local build from a committed, pushed and successfully deployed website. Tests catch implementation mistakes but cannot certify chemical realism.

15. Parameter and reference registry

Glycerol-inspired: mass 92.094 Da; illustrative dipole 2.6 D; sigma 0.48 nm; molecular epsilon 6.0 kJ/mol. Density anchor 1.2613 g/cm³ at 293.15 K; pressure unspecified. Source: CRC Handbook, 88th ed., via PubChem/HSDB.

glycerol: no vaporization enthalpy value in the curated registry.

Ethylene glycol-inspired: mass 62.068 Da; illustrative dipole 2.3 D; sigma 0.43 nm; molecular epsilon 4.5 kJ/mol. Density anchor 1.1135 g/cm³ at 293.15 K; pressure unspecified. Source: CRC Handbook, 91st ed., via PubChem/HSDB.

ethylene_glycol: vaporization enthalpy 62.4 kJ/mol at 345.0 K; pressure unspecified kPa. Petitjean, Reyes-Perez et al. (2010), via NIST WebBook; 62.4 +/- 4.0 kJ/mol at 345 K, based on 307-384 K data.

DMSO-inspired: mass 78.133 Da; illustrative dipole 3.96 D; sigma 0.46 nm; molecular epsilon 5.0 kJ/mol. Density anchor 1.1 g/cm³ at 293.15 K; pressure unspecified. Source: Merck Index (2013), via PubChem/HSDB; relative density 20/4 C.

dmso: vaporization enthalpy 51.7 kJ/mol at 320.0 K; pressure unspecified kPa. Stephenson and Malanowski (1987), NIST WebBook; based on 305-464 K data.

Propylene glycol-inspired: mass 76.095 Da; illustrative dipole 2.3 D; sigma 0.46 nm; molecular epsilon 5.0 kJ/mol. Density anchor 1.0361 g/cm³ at 293.15 K; pressure unspecified. Source: CRC Handbook, 88th ed., via PubChem/HSDB.

propylene_glycol: no vaporization enthalpy value in the curated registry.

Formamide-inspired: mass 45.041 Da; illustrative dipole 3.73 D; sigma 0.36 nm; molecular epsilon 4.5 kJ/mol. Density anchor 1.1334 g/cm³ at 293.15 K; pressure unspecified. Source: Merck Index (1996), via PubChem/HSDB; relative density 20/4 C.

formamide: no vaporization enthalpy value in the curated registry.

Benzyl alcohol-inspired: mass 108.14 Da; illustrative dipole 1.7 D; sigma 0.5 nm; molecular epsilon 5.5 kJ/mol. Density anchor 1.0419 g/cm³ at 297.15 K; pressure unspecified. Source: CRC Handbook, 94th ed., via PubChem/HSDB.

benzyl_alcohol: vaporization enthalpy 50.48 kJ/mol at 478.46 K; pressure 101.325 kPa. CRC Handbook, 94th ed., via PubChem/HSDB: normal boiling point 205.31 C.

Available Hamiltonians: coulomb_lj: Partial-charge Coulomb + LJ; lj: Lennard-Jones only;
coulomb_wca: Partial-charge Coulomb + WCA; wca: WCA repulsion only.

16. Sources and further reading

OpenMM NonbondedForce: Coulomb, LJ, PME, switching and dispersion correction
<https://docs.openmm.org/latest/api-python/generated/openmm.openmm.NonbondedForce.html>

OpenMM LangevinMiddleIntegrator <https://docs.openmm.org/latest/api-python/generated/openmm.openmm.LangevinMiddleIntegrator.html>

OpenMM MonteCarloBarostat <https://docs.openmm.org/latest/api-python/generated/openmm.openmm.MonteCarloBarostat.html>

Weeks, Chandler and Andersen (1971): repulsive reference separation
<https://doi.org/10.1063/1.1674820>

Steinhardt, Nelson and Ronchetti (1983): bond-orientational order
<https://doi.org/10.1103/PhysRevB.28.784>

LAMMPS orientorder/atom: definitions and neighbor choices
https://docs.lammps.org/compute_orientorder_atom.html

LAMMPS MSD: unwrapping and reference choices https://docs.lammps.org/compute_msd.html

GROMACS energy: ensemble-specific fluctuation properties and caveats
<https://manual.gromacs.org/current/onlinehelp/gmx-energy.html>

SciPy PCHIP: shape-preserving interpolation
<https://docs.scipy.org/doc/scipy/reference/generated/scipy.interpolate.PchipInterpolator.html>

glycerol: density (293.15 K) <https://pubchem.ncbi.nlm.nih.gov/compound/753#section=Density>

ethylene_glycol: density (293.15 K)
<https://pubchem.ncbi.nlm.nih.gov/compound/174#section=Density>

ethylene_glycol: vaporization enthalpy
<https://webbook.nist.gov/cgi/cbook.cgi?ID=C107211&Mask=4>

dmso: density (293.15 K) <https://pubchem.ncbi.nlm.nih.gov/compound/679#section=Density>

dmso: vaporization enthalpy <https://webbook.nist.gov/cgi/cbook.cgi?ID=C67685&Mask=4>

propylene_glycol: density (293.15 K)

<https://pubchem.ncbi.nlm.nih.gov/compound/1030#section=Density>

formamide: density (293.15 K) <https://pubchem.ncbi.nlm.nih.gov/compound/713#section=Density>

benzyl_alcohol: density (297.15 K)

<https://pubchem.ncbi.nlm.nih.gov/compound/244#section=Density>

benzyl_alcohol: vaporization enthalpy

<https://pubchem.ncbi.nlm.nih.gov/compound/244#section=Heat-of-Vaporization>